(iii) and (iv), we can obtain individually, the geometrical areas of the regions corresponding to the subintervals.

Hence the geometrical area A of the region bounded by the graph of $x = f(y)$, the y-axis, the lines $y = c$ and $y = d$ is given by

$$A = \left|\int_c^{a_1} f(y)dy\right| + \left|\int_{a_1}^{a_2} f(y)dy\right| + \cdots + \left|\int_{a_k}^{d} f(y)dy\right|.$$

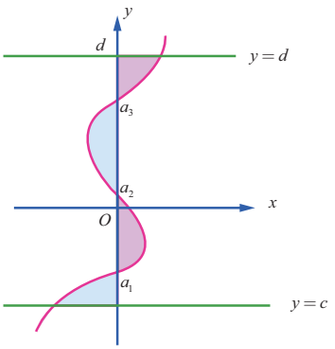


Fig. 9.13

For instance, consider the shaded region in Fig. 9.13. Here, B_1, B_2, B_3 and B_4 denote geometric areas of the individual parts. Then, the total area B of the region bounded by the curve $x = f(y)$, $y-$axis and the lines $y = c$ and $y = d$ is given by

$$\begin{aligned} B &= B_1 + B_2 + B_3 + B_4 \\ &= \left|\int_c^{a_1} f(y)dy\right| + \int_{a_1}^{a_2} f(y)dy + \left|\int_{a_2}^{a_3} f(y)dy\right| + \int_{a_3}^{d} f(y)dy. \end{aligned}$$

Example 9.47

Find the area of the region bounded by the line $6x + 5y = 30$, $x-$axis and the lines $x = -1$ and $x = 3$.

Solution

The region is sketched in Fig. 9.14. It lies above the $x-$axis. Hence, the required area is given by

$$\begin{aligned} A &= \int_{-1}^{3} y\,dx = \int_{-1}^{3}\left(\frac{30-6x}{5}\right)dx = \left(\frac{30x-3x^2}{5}\right)_{-1}^{3} \\ &= \left(\frac{90-27}{5}\right) - \left(\frac{-30-3}{5}\right) = \frac{96}{5}. \end{aligned}$$

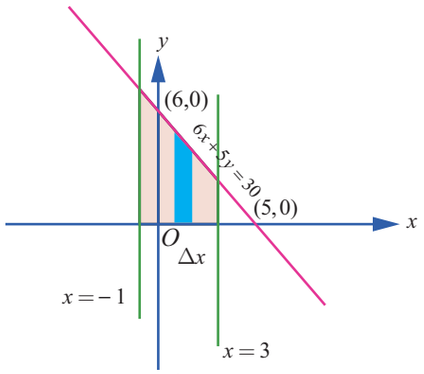


Fig. 9.14

Example 9.48

Find the area of the region bounded by the line $7x - 5y = 35$, $x-$axis and the lines $x = -2$ and $x = 3$.

Solution

The region is sketched in Fig. 9.15. It lies below the $x-$axis. Hence, the required area is given by

$$A = \left|\int_{-2}^{3} y\,dx\right| = \left|\int_{-2}^{3}\left(\frac{7x-35}{5}\right)dx\right|$$

$$= \frac{1}{5}\left|\left(7\left(\frac{x^2}{2}\right) - 35x\right)_{-2}^{3}\right|$$

$$= \frac{1}{5}\left|\left(\left(\frac{63}{2}\right) - 105\right) - (84)\right| = \frac{63}{2}.$$

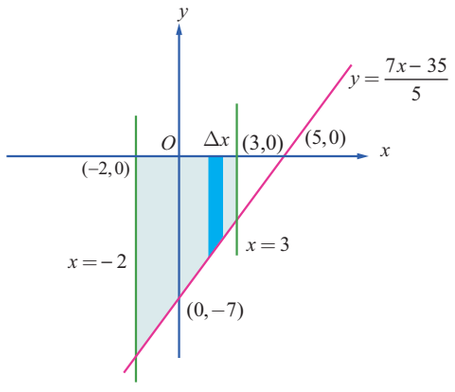


Fig. 9.15

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